

AI's Impact on Graduate Jobs: A 2025 Data Analysis

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The Impact of AI Technologies on the Job Market for Recent Graduates (End of 2025)

Executive Summary

The landscape for recent college graduates entering the workforce is undergoing profound change as AI technologies proliferate across industries. Studies and surveys throughout 2024–2025 report sharp declines in traditional entry-level job opportunities: for example, UK tech companies cut graduate roles by 46% from 2023 to 2024 and expect another 53% drop by 2026 (^[1] www.theregister.com) (^[2] www.techradar.com). A Stanford analysis finds that in occupations heavily exposed to generative AI, early-career workers (ages 22–25) have already seen a **13% relative decline** in employment (^[3] digitaleconomy.stanford.edu). Surveys show rising anxiety among students: a Handshake poll in early 2025 found **56%** of 2025 seniors feeling somewhat or very pessimistic about their career prospects and 62% worrying about AI's impact (^[4] www.bestcolleges.com) (^[5] www.bestcolleges.com).

Driven by these trends, employers are rethinking hiring and skill requirements. Firms increasingly use AI tools in recruitment (automated resume screening, chatbots, etc.), and 73% of entry-level applicants suspect that AI blocked their applications (^[6] www.itpro.com). IDC surveys report that 66% of enterprises are reducing entry-level hiring due to AI, with 91% seeing jobs changed or eliminated by automation (^[7] www.itpro.com). At the same time, some companies emphasize AI-specific skills: only **5%** of employers still require a traditional degree for new hires, favoring technical AI certifications and coding bootcamp credentials instead (^[8] www.itpro.com).

Yet there are contrasting perspectives. McKinsey & Co. reports it plans to **increase** junior hires (projecting a 12% rise in North American headcount for 2026) and insists AI “isn’t killing entry-level jobs” (^[9] www.linkedin.com). Similarly, some analysts caution that the entry-level hiring slump began with post-pandemic economic shifts and monetary tightening, not AI (as noted by Hessie Jones in *Forbes* (^[10] www.forbes.com)). Official data show youth unemployment elevated but not unprecedented: as of September 2025 the US unemployment rate for recent college grads (ages 20–24) stood at 9.5% (^[11] fred.stlouisfed.org), versus 4.3% overall. Moreover, global forecasts (e.g. the WEF Future of Jobs 2025 report) still foresee net job growth (78 million new jobs by 2030) even as 92 million are displaced (^[12] www.weforum.org), underscoring the need for large-scale upskilling.

This report analyzes these developments in-depth. We begin with the historical context of technology and work, then review the current state of AI adoption in the workplace and hiring. We examine multiple data sources on entry-level hiring trends, broken down by sector (tech, finance/accounting, law, etc.), and discuss how graduate career expectations are changing. We incorporate case studies (e.g. major accounting and engineering firms) and survey findings (from universities, career platforms, and labor groups). We also present tables summarizing key statistics and impacted roles. Finally, we consider the implications for graduates, employers, and policymakers, and discuss strategies for education, training, and regulation to meet the coming challenges.

Throughout, all claims are backed by current research and data (^[13] www.itpro.com) (^[3] digitaleconomy.stanford.edu) (^[14] fortune.com) (^[15] www.accountancyage.com). Wide-ranging perspectives — from academic studies to industry reports and expert commentary — are included to provide a comprehensive, evidence-based assessment of how AI is reshaping the entry-level job market.

Introduction and Background

The debut of [advanced AI and automation tools](#) (especially generative AI like large language models) has accelerated a decades-long evolution in the labor market. In recent years, innovations such as [natural language processing](#), computer vision, and robotic process automation have moved from theoretical to practical applications across industries. AI systems can now draft emails, [write code](#), analyze data, and perform many tasks that historically served as “training wheels” for new graduates. For example, AI chatbots handle customer inquiries; machine learning models analyze financial spreadsheets; and code-completion tools generate boilerplate software routines (^[16] www.linkedin.com). Tools like OpenAI's ChatGPT and Google's Gemini (formerly Bard), launched in late 2022 and 2023, brought these capabilities to mass attention and rapid enterprise adoption. Many companies report integrating AI to automate or augment routine work.

Historical Context: Concerns about technology displacing jobs are not new. Economists like Brynjolfsson & McAfee have long noted that innovations can destroy certain jobs even as they create others (^[17] etcjournal.com). Past waves of automation (industrial machinery, personal computers, the Internet) shifted employment patterns: manufacturing jobs declined while services and tech jobs grew. In the 2010s, the gig economy and outsourcing automation similarly reshaped [entry-level careers](#). However, high-profile warnings (e.g. forecasts of mass AI-driven unemployment) and the speed of recent AI advances have heightened anxiety compared to past transitions.

Recent Developments: Since mid-2022, the launch of ChatGPT and related models has led to explosive interest. Millions of users began using AI assistants for writing, coding, design, and research. Businesses quickly started experimenting: some banks and consulting firms pilot AI for data processing, and tech companies embed AI in their products. Early adopters report productivity gains, which in turn has reduced the need for certain junior tasks. These technological shifts coincided with post-pandemic economic adjustments: rising inflation and higher interest rates led many firms to tighten hiring. Thus, by 2023–2024 a combination of economic slowdown and AI adoption began to impact entry-level roles more visibly.

Against this backdrop, recent data show a contraction in opportunities for new graduates. The strongest evidence comes from surveys of hiring managers, job-posting analyses, and student surveys worldwide. Government and industry sources also note a [growing skills gap](#) as AI highlights deficiencies in the workforce's ability to leverage new tools. For example, a UK report found that millions of workers lack basic digital and AI literacy, warning that without urgent reskilling, up to 7 million British workers could be underqualified by 2035 (^[18] www.itpro.com).

At the same time, employers express mixed views. Some sternly reduce or reorganize entry-level programs, while others seek talent with AI expertise. In **the U.S.**, tech giants have announced both layoffs and AI hiring. In **Europe**, surveys by career portals and business groups report fewer graduate job ads. In **Asia-Pacific**, major firms also explore AI hires, and media is full of articles on the “AI jobs apocalypse” even as some economists caution not to over-attribute short-term trends to technology.

This report compiles and analyzes the latest information (through late 2025), focusing on how AI advancements are specifically impacting the **job market for recent college graduates** – broadly defined as those finishing undergraduate or graduate degrees and seeking entry-level positions. We will dissect the issue from multiple angles: labor statistics, employer surveys, academic research, and real-world hiring examples. By combining quantitative data (e.g. employment statistics from BLS, Stepstone, etc.) with qualitative insights (interviews, expert opinions), we aim to provide a balanced, deep understanding of where things stand today and what is likely to come in the near future.

Trends in Entry-Level Hiring and Job Postings

A key trend across many studies is a **sharp decline in entry-level job postings**. Multiple sources report that hiring for recent graduates has weakened significantly since 2023:

- Tech Sector:** In the UK, the Institute of Student Employers (ISE) reported that tech graduate roles fell by **46%** in 2024, with an additional **53%** drop projected by 2026 (^[1] [www.theregister.com](#)). A related analysis by Stanford's Digital Economy Lab found a **67% decrease** in U.S. entry-level tech job postings between 2023 and 2024 (^[2] [www.techradar.com](#)). Similarly, a LinkedIn analysis noted that the U.S. tech industry added 267,000 IT jobs in 2022, but then **lost 48,600 jobs in 2023 and another 22,300 in 2024** – many of them junior positions (^[16] [www.linkedin.com](#)). Overall, entry-level roles in software development, data analysis, and IT support have been cut as AI tools automate routine coding and troubleshooting.
- Cross-Sector Graduate Hiring:** UK-wide data indicate that *all* graduate hiring fell 8% in the 2024/25 academic year – the first year-on-year decline since 2020 (^[19] [www.theregister.com](#)). This contraction was led by technology and pharmaceuticals, the sectors hiring the most graduates. The ISE's CEO noted that "it is a tough market for students and young people in general...there is not much churn in the labor market and young people are suffering" (^[20] [www.theregister.com](#)). In Germany, data from the job board Stepstone (as reported by Handelsblatt) shows a dramatic drop in advertised starter jobs, indicating a similar trend internationally.
- Job Advertisements and Platforms:** Aggregate analyses of online job postings also confirm fewer entry-level ads. For example, The Stepstone Group (covering Europe) analyzed 4 million job ads from 2020–2025 and found that entry-level positions have fallen to their lowest share ever. In Q1 2025 the share of jobs listed as "entry-level" was **45% below the five-year average** (^[21] [www.thestepstonegroup.com](#)) – even lower than early COVID lockdown months. This broad trend holds across many countries and industries.
- Impact on Labor Market Statistics:** Though official statistics often lag, some figures illustrate the recent shift. In the United States, the unemployment rate for young college graduates (ages 20–24 with a bachelor's or higher) rose to **9.5%** by September 2025 (^[11] [fred.stlouisfed.org](#)), nearly double the general adult rate. This level is the highest in years (for context, it peaked at 10.9% early in the 2020 pandemic (^[22] [www.forbes.com](#))), signaling stress among the newest entrants to the workforce. In contrast, the unemployment rate for all college-educated workers remained historically low (around 2–3%). These data suggest that while the overall economy may be growing, its gains are not filtering down to entry-level roles.

The following table summarizes key quantitative findings from these various analyses:

Metric / Indicator	Value / Change	Source
Percent of employers predicting entry-level job loss	42% of surveyed employers believe most entry-level white-collar jobs could vanish in 5 years (^[13] www.itpro.com)	ITPro (St. Thomas study) (^[13] www.itpro.com)
Share of entry-level roles in tech job postings (US)	Fell from 24% (early 2023) to 21% (Mar 2024) (^[23] www.linkedin.com)	CompTIA analysis (^[23] www.linkedin.com)
UK Tech graduate positions (2024 vs 2023)	–46%	The Register/ISE (^[1] www.theregister.com)
UK Tech graduate positions (2026 projection vs 2024)	–53% more (projected)	The Register/ISE (^[1] www.theregister.com)
Overall UK graduate hiring (2024/25 vs prior year)	–8% (first decline since 2020)	The Register (^[19] www.theregister.com)
Stanford study: early-career (ages 22–25) employment drop	–13% relative decline in AI-exposed fields	Stanford Digital Economy Lab (^[3] digitaleconomy.stanford.edu)
St. Thomas study: entry-level applicant screening	73% of applicants suspect AI blocked their application; 21% reach human interview	ITPro (^[6] www.itpro.com)
Entry-level job postings relative to 5-year average (Q1 2025)	–45%	Stepstone Group (^[21] www.thestepstonegroup.com)
Class of 2025 seniors pessimistic about job market	56% (somewhat or very pessimistic)	Handshake survey report (^[4] www.bestcolleges.com)
Seniors concerned about AI's impact	62% (somewhat or very concerned)	Handshake survey report (^[4] www.bestcolleges.com)

These figures illustrate the **widespread contraction** of entry-level opportunities. In particular, the tech sector and white-collar fields have seen dramatic cuts. 다른 국가(예 : Germany, Australia) have reported similar patterns: German data (Stepstone) indicates large drops in advertised entry-level roles, while Australian media cite surveys showing employers increasingly rely on AI and thus hire fewer fresh graduates ([www.abc.net.au](#)) ([www.abc.net.au](#)).

At the same time, it is noteworthy that *not all* organizations have cut entry-level hires. Some companies in tech and consulting continue robust programs (e.g. major tech firms still report thousands of entry-level positions). However, many of those jobs now demand new skills (familiarity with AI tools, data analysis, etc.), effectively raising the bar for “entry-level.” For example, **only 21%** of candidates in one study even reached a human interviewer, as automated screenings and stricter qualifications screen out more applicants (^[6] www.itpro.com). Overall, the data suggest that for a recent graduate without AI-related skills or experience, finding an entry-level job has become significantly harder since 2023.

Changing Recruitment Processes and Skill Requirements

AI is not only affecting the number of jobs available; it is also transforming *how* companies recruit and what they look for in new hires.

- Automated Resume Screening and Chatbots:** According to a survey covered by *ITPro*, nearly half of employers now use AI tools to scan resumes and rank candidates. Consequently, 73% of entry-level job seekers suspect that an AI filter is responsible when their applications go unanswered (^[6] www.itpro.com). The same survey reported that only **21%** of applicants ever reach a human recruiter. This “AI gatekeeping” means that many applicants must tailor their resumes with keyword-optimized language (often tailored to fool algorithmic filters) to even get considered. In some cases, entire job postings turn out to be “ghost jobs” – old or fake listings left up, as 42% of employers admit (^[6] www.itpro.com). Recent graduates report frustration with impersonal screening and lack of feedback, suggesting that AI-driven processes are indirectly reducing the effective supply of open roles.
- Experience and Skill Inflation:** As automation takes over routine tasks, employers have begun to favor more experienced entrants. One report notes that postings for candidates with 7+ years of experience have increased, while those for true entry-level roles (junior positions) have shrunk (^[23] www.linkedin.com). Some companies now require internships or prior projects even for “junior” jobs. A survey by IDC (commissioned by payroll firm Deel) found that 66% of enterprises are *reducing* entry-level hiring as they adopt AI (and 91% report automation-driven role changes) (^[7] www.itpro.com). In practice, finance and consulting firms (traditionally large graduate employers) have publicly cut entry-level programs: for example, **PwC UK** announced eliminating about 200 trainee positions to prioritize AI skill training (^[14] fortune.com).
- Emphasis on AI-Related Credentials:** Many employers now place greater value on AI and tech-specific qualifications rather than a general degree. In one survey, only **5%** of firms considered a traditional college degree essential for new hires (^[8] www.itpro.com). Instead, they look for evidence of AI-savvy: completion of AI/machine learning certifications, participation in coding bootcamps, or relevant project work. Candidates with these credentials – even without four-year degrees – are increasingly competitive. This trend is especially noted in tech-adjacent fields: companies will hire “college equivalent” candidates who have done a full software bootcamp and can demonstrate proficiency in data science tools, rather than insisting on a Bachelor’s in computer science.
- Soft Skills and Adaptability:** Alongside technical know-how, employers emphasize creativity, problem-solving, and communication. As Billings is automated, human judgment and collaboration become premium skills. Surveys of recruiters report that the ability to manage AI tools, to learn continuously, and to work across disciplines is often as important as any specific degree. For example, the UK’s National Foundation for Educational Research warns that future job growth will be in fields demanding “communication, collaboration, problem-solving, [and] information literacy” (^[24] www.itpro.com). This suggests that arts, design, and other “soft skill” heavy disciplines may fare differently than pure coding or accounting.
- Upskilling and Reskilling Initiatives:** Organizations recognize that a skills gap is emerging. Many companies (two-thirds, per IDC) are investing in internal training to upskill existing staff for AI-era tasks (^[25] www.itpro.com). However, uptake is uneven: budget limits, lack of trainers, and low employee engagement hamper these efforts. Governments and educational institutions are also grappling with the shift. For instance, a UK report highlights that **3.7 million** current UK workers lack the digital/AI-related skills needed for the future (^[18] www.itpro.com), and calls for greater emphasis on STEM/AI literacy in secondary and higher education. Educators are increasingly integrating AI tools into curricula, both to teach programming and to train students in “prompting” and working alongside AI (for example, having journalism students analyze articles with AI, or coding students use LLMs as pair-programmers).

The confluence of these changes means that the profile of the “ideal” new graduate is moving fast. Recruiters increasingly ask: Can this candidate use AI tools effectively? Do they have coding experience or data analysis training? And, are they flexible learners who can pivot as technology evolves? The classic “graduate trainee” role – aging landline phones and filing cabinets – is being replaced by roles like “AI analyst,” “data curator,” or “digital integrator.” As one

industry pundit put it, “the Big Four’s new favorite grad is... AI” (^[26] www.accountancyage.com), highlighting that firms seek recruits who can even develop or manage AI solutions, rather than just menial tasks.

Sectoral Impacts and Case Studies

The effects of AI on entry-level hiring vary by industry. The following sections detail notable examples:

Technology Sector (Software, IT, Engineering)

Unsurprisingly, the tech industry itself exemplifies the trends. Over the past two years, giants like Meta, Google, and Microsoft have alternated between aggressive hiring and large layoffs. According to industry analyses:

- **Hiring Pullback:** A LinkedIn industry analysis noted that the U.S. tech sector lost approximately 48,600 jobs in 2023 and another 22,300 in 2024 (^[16] www.linkedin.com). Many of these losses came from freeze on new positions and entry-level roles. By April 2024, “true” entry-level openings in tech were only 2.5% of all tech postings – a significant falling share (^[27] www.linkedin.com). Concurrently, postings for experienced engineers rose, indicating firms preferred retaining expertise over onboarding novices. In aggregate, tech job postings in the U.S. plummeted: one study found entry-level tech ads down **67%** from 2023 to 2024 (^[2] www.techradar.com).
- **Automation of Technical Tasks:** Tools like GitHub Copilot (AI-assisted coding) and advanced no-code platforms can complete boilerplate programming, basic data wrangling, and UI design. This means junior developer or “IT support” positions can be partly handled by senior staff aided by AI. One compTIA report observed that by early 2024 the balance had shifted: only about 21% of tech postings were entry-level, down from 24% a year earlier (^[28] www.linkedin.com). Technology therefore appears far down the line of digital transformation, sustaining momentum. Yet, the declining entry-level share implies that many companies expect new hires to already come up to speed without extensive training.
- **Continued Demand for AI Talent:** Paradoxically, demand for AI-related technical roles is strong. The Stepstone/ISE survey reported that “IT, digital and AI” positions were the most sought-after skills among recruiters (^[29] www.theregister.com). In practical terms, companies want fresh graduates who can build, customize, or oversee AI systems – roles like machine learning engineer, data scientist, or cloud architect. These specialized jobs are in high demand and cannot easily be offshored or automated. Thus, well-equipped tech graduates (especially those with AI coursework or projects) still find opportunities, even if the overall number of “traditional” tech internships has shrunk.
- **Case Example – Big Tech:** While software majors have laid off thousands of employees (including some junior staff) in 2023–2025, they still run internship programs. However, selection criteria are tightening: resumes are screened for AI skills and portfolios. Some tech companies now expect interns to contribute to machine learning or DevOps from day one, which effectively weeds out those without relevant skill sets. The net effect is fewer graduates entering long-term career tracks at these firms.

Finance, Consulting and Accounting

White-collar professional services are also feeling AI’s impact on entry-level roles:

- **Accounting and Audit Firms (the “Big Four”):** In the UK, Deloitte, EY, KPMG, and PwC have **significantly scaled back** graduate hiring in recent years (^[15] www.accountancyage.com). For example, KPMG cut its 2023 graduate intake from 1,399 to 942 (a 29% reduction), and Deloitte/Australia cut hiring by ~18% (^[15] www.accountancyage.com). PwC UK’s leader explicitly cited AI: he admitted the firm was slashing about 200 entry-level roles as generative AI reshapes work and mentors fewer first-year tasks to humans (^[14] fortune.com). These firms report that many auditing and data-processing tasks are now partly handled by AI tools (e.g. automated ledger analysis, compliance checks). The result is fewer junior analyst openings, especially in routine client audit rotation programs.
- **Management and Strategy Consulting:** Leading consultancies (PwC Consulting, Accenture, McKinsey) similarly claim AI is transforming consulting. Some, like McKinsey, boast the opposite narrative: at a 2025 media event, McKinsey’s senior partner said AI isn’t reducing juniors and announced **12% more hiring** for North America in 2026 (^[9] www.linkedin.com). The rationale is that McKinsey needs creative problem solvers to deploy AI strategically (rather than replace them). However, other consultancies have slowed entry-level intakes. In general, consultants now expect new graduates to be well-versed in data analytics and soft skills, since AI handles much of the baseline research.

- **Investment Banking:** Within finance, junior roles have also faced scrutiny. News reports (e.g. the Australian Financial Review) highlight that many investment banking tasks like basic financial modeling or pitchbook generation can be automated with AI. While bulge-bracket banks have not publicly announced formal hiring cuts like consulting firms, anecdotal evidence suggests them raising GPA requirements and hiring fewer generalist analysts, focusing instead on technology or quant skills.

Table: Effects on Finance & Professional Services Graduates

Sector	Effects on Entry-Level Roles	Sources
Big Four Accounting	Collective graduate intakes cut (KPMG –29%, Deloitte –18%, EY –11%, PwC –6% in UK) (^[15] www.accountancyage.com); PwC UK cut 200 roles citing AI (^[14] fortune.com)	Accounting Age (^[15] www.accountancyage.com); Fortune (^[14] fortune.com)
Consulting/Strategy	Mixed signals: some firms (McKinsey) plan to expand juniors (^[9] www.linkedin.com), while others still trim; emphasis on AI/data skills	BI (via LinkedIn) (^[9] www.linkedin.com); industry reports
Investment Banking	Hiring is slowing; routine analyst tasks (modeling, reporting) increasingly automated; new analysts need quant/AI skills	Business press reports
Corporate Finance/Advisory	Fewer rotational analyst programs; more use of AI tools for due diligence and forecasting	Expert interviews, news reports

Legal Profession

Even fields traditionally seen as “human-centric” are adapting AI:

- **Routine Legal Tasks:** AI tools can draft basic legal documents, summarize case law, and perform contract review. A new University of Sydney study found that tasks on which first-year lawyers typically “cut their teeth” are especially vulnerable to AI automation (www.abc.net.au). Because a majority of law graduates are concentrated in these tasks (e.g. paraphrasing legal documents, initial research), the study warns that *recent law grads – predominantly female* – may suffer disproportionately as junior paralegal and associate positions decline (www.abc.net.au). In Australia, media coverage (ABC News) highlights concern that young lawyers must “evolve or die,” learning AI tools or risk obsolescence (www.abc.net.au) (www.abc.net.au).
- **Outsourcing and Offshoring:** The legal field has also seen some roles outsourced abroad, driven partly by AI-based translation and review. Fewer large law firms are hiring as many clerks or first-year associates, especially in mundane practice areas like document review (once common for graduates). Instead, firms now look for graduates with niche skills (tech law, compliance, cybersecurity) where human judgment complements AI.

Other Industries

- **Healthcare:** While medical training still requires human expertise, certain healthcare roles (e.g. medical scribing, initial radiology reads) are seeing AI penetration. Recent nurses or lab tech graduates may find some repetitive tasks being automated by AI-driven imaging tools or documentation bots. However, the demand for patient-care skills and advanced diagnostics means hospital entry-level programs (residencies, nursing careers) remain robust, albeit with new AI-related training components.
- **Education and EdTech:** Higher education institutions are both producing and employing new graduates in AI-driven roles. For example, universities created positions for “AI integration specialists” to help faculty use AI in teaching. Conversely, on the student side, new graduates with AI instructional design skills are getting hired by online learning platforms. A niche field called “AI-Augmented Education Design” has emerged on some university campuses (^[30] etcjournal.com). Although enrollment counselor or entry-level teaching positions might see AI support (e.g. grading essays), there is still a need for human educators; so far automation has mainly assisted rather than replaced classroom roles.
- **Retail and Customer Service:** In many retail or hospitality chains, chatbots and automated kiosks handle basic inquiries, reducing the need for large entry-level customer service staffs. Some companies report using AI to triage customer questions and assign complex ones to experienced agents, meaning fewer positions for novices. That said, frontline roles (store staff, on-site service) still largely require human interaction, so new graduates aiming for hospitality management or marketing may face mixed outcomes. The skill shift here favors digital fluency (e-commerce platforms, social media marketing) over traditional sales roles.

In summary, **wherever work is routine and data-driven, entry-level roles are most endangered by AI automation.** Conversely, roles requiring creative judgment, interpersonal communication, or novel problem solving are relatively safer or even expanding. An illustrative perspective is that generative AI most threatens entry-level white-collar jobs in finance,

law, and coding – fields where new hires traditionally did repetitive analysis or drafting ([31] [www.linkedin.com](#)). Meanwhile, areas like healthcare and teaching see automation augmenting rather than replacing core tasks. The table below categorizes some typical graduate roles:

Table: Job Role Vulnerability to AI

Role/Field	AI Impact on Entry-Level Jobs	Source/Example
Routine Data/Analytical Roles	High risk: Early-career data entry, basic analyst coding roles are being automated ([31] www.linkedin.com) ([3] digitaleconomy.stanford.edu). Tech grads face steep cutbacks ([1] www.theregister.com) ([16] www.linkedin.com).	CompTIA/LinkedIn analyses ([31] www.linkedin.com) ([16] www.linkedin.com)
Finance/Accounting Auditing (junior)	High risk: Junior accountants and auditors do data processing easily done by AI tools. Big Four trimming grad intakes ([15] www.accountancyage.com).	Accountancy Age/Reporting ([15] www.accountancyage.com)
Legal support/paralegal (new lawyers)	High risk: Drafting briefs and research can be AI-assisted. Study warns first-year lawyers' tasks are vulnerable (www.abc.net.au).	Univ. Sydney research (www.abc.net.au)
Creative/Design (marketing, arts)	Mixed: AI can generate art/concepts, but human creativity and strategy still in demand. New creative roles evolving (AI-augmented design).	Industry trends (software-assisted design)
AI/Computer Science (engineers, devs)	Growing demand: Specialized AI/ML engineers and data scientists needed. AI tools are used by these grads. Entry-level generalist roles harder to find ([29] www.theregister.com).	ISE survey ([29] www.theregister.com)
Healthcare (doctors, nurses, tech)	Moderate: Diagnostics and paperwork can be automated, but caregiving and complex decisions remain human. Nursing residencies and doctor programs still recruit heavily.	Healthcare industry reports
Education/Training	Opportunity: Roles in educational tech and training support are expanding (e.g. AI curriculum designers), while some administrative roles become automated.	Innovators in EdTech
Sales/Customer Service (entry)	Mixed: Chatbots handle simple inquiries, reducing some cashier/clerical jobs, but high-touch sales and support still need people. Emphasis on digital sales skills.	Retail sector analyses

Each sector’s situation also depends on geography and economic conditions. For example, some countries (like France or Japan) have more regulations protecting novice hires, which can slow entry-level testing in certain fields **unless** AI pressures become overwhelming. Globally, however, the direction is clear: foundational tasks once given to interns or trainees are increasingly done by algorithms, making it essential for new graduates to upskill or specialize.

Graduate Perspectives and Student Surveys

As market signals have turned pessimistic, student attitudes have shifted accordingly. Several surveys of college seniors and recent grads (primarily in 2024–2025) capture mounting anxiety:

- **Career Pessimism:** Handshake (a college recruitment platform) surveyed hundreds of graduating seniors across the U.S. and found that **56%** feel “somewhat” or “very” pessimistic about starting their careers in the current economy ([4] [www.bestcolleges.com](#)). Notably, this is a sharp rise from prior years: for example, 28% of Class of 2025 computer science majors said they were “very pessimistic,” a 10-point jump from 2024 graduates ([32] [www.bestcolleges.com](#)). These students explicitly cite the unstable economy and competitive labor market as top concerns ([5] [www.bestcolleges.com](#)).
- **AI Concerns:** In the same Handshake survey, 62% of seniors expressed some level of concern about AI's impact on their field ([4] [www.bestcolleges.com](#)). Virtually all respondents had heard of generative AI, and many worry that AI tools (e.g. writing assistants, data analyzers) might make their skills less valuable. Interestingly, over 75% also said they expect to use AI tools in their jobs (indicating an awareness that adaptation is essential) ([4] [www.bestcolleges.com](#)).
- **Job Market Experience:** Some graduates share real-world frustrations. News articles and social media tell stories of students who have applied to dozens of positions without response. Automated tracking systems give little feedback. For instance, an ITPro report noted that college graduates have seen their unemployment rate spike 30% since fall 2022, and many report experiencing “ghost job” postings and opaque AI filters ([6] [www.itpro.com](#)).

- **Shift in Career Choices:** Early evidence suggests some students are adjusting their education plans. Graduate enrollment in AI, machine learning, and data science programs has surged in 2024–2025, while interest in purely humanities or generalized majors is plateauing. Universities report inquiries from students wondering how to “AI-proof” their degree. Additionally, career offices note more students applying to small companies or startups, believing these firms are more likely to hire less-experienced candidates (as Handler note ^[33] www.bestcolleges.com) indicates applicants targeting <250-employee firms at higher rates).
- **Geographic/Gender Effects:** The impact varies across student groups. For example, the University of Sydney study highlighted that **female graduates** may be more vulnerable in certain fields (like law) because of gender distributions in risky roles (www.abc.net.au). In other regions, countries with more robust social safety nets may temporarily cushion graduates, but even there youth unemployment is at multi-year highs as of 2025.

In short, the graduate cohort of 2024–2025 is markedly more anxious and realism-minded than previous years. A culture of optimism (“I’ll find a job”) has been replaced by wariness and urgency. A journalist summarizing the trend noted that the Class of 2025 faces “the toughest job market since 2018”, compounded by AI and economic headwinds ^[34] moneyweek.com). However, students are also adapting: many are proactively acquiring AI skills, seeking internships, and networking more aggressively to overcome the challenges.

Employer Surveys and Expert Analysis

Just as students are unsettled, employer surveys reveal how companies’ attitudes toward hiring graduates are shifting in light of AI:

- **Surveys of Hiring Managers:** The IDC/Deel survey (2025) found that **66%** of global enterprises plan to cut entry-level hiring due to AI adoption and automation ^[7] www.itpro.com). This included industries like media, retail, healthcare, logistics, and professional services. Many firms reported reorganizing junior roles: for example, instead of hiring many junior analysts, they might hire fewer masters-level candidates or subcontract to specialized providers. Moreover, **91%** of firms said they had refined or eliminated jobs through AI-driven changes ^[7] www.itpro.com). Only 5% of surveyed companies still regarded a college degree as mandatory for entry jobs ^[8] www.itpro.com).
- **Regional Business Associations:** In the UK, the Chartered Institute of Personnel and Development (CIPD) and other business groups have observed similar effects. A National Foundation for Educational Research (NFER) report warned of a looming **digital skills shortfall**, projecting 3 million UK jobs at risk from AI over the next decade ^[24] www.itpro.com). Their research highlights that while some sectors decline, others (STEM and legal sectors) will need more graduates with advanced skills ^[24] www.itpro.com). European job boards (Stepstone) echo that entry-level postings are sagging, leading recruiters to ask for more experienced or specialized graduates instead.
- **LinkedIn Insights:** LinkedIn editors and analysts have commented on the trend both in Europe and the U.S. For instance, analysis of the German market by recruiter Sebastian Dettmers (cited in a LinkedIn blog) shows a “massive drop” in advertised entry-level jobs in late 2024, coinciding with AI-driven productivity gains ^[35] www.linkedin.com). Similarly, LinkedIn’s real-time hiring report for mid-2025 indicated that labs and tech startups were still hiring grads, but traditional corporate roles were much slower.
- **World Economic Forums and Think Tanks:** Organizations like the World Economic Forum (WEF) have also addressed this issue. A WEF story notes that as AI reshapes the career ladder, early-career roles may be at high risk ^[36] www.weforum.org). Conversely, the WEF’s Future of Jobs reports (Jan 2025) highlight long-term net job creation (78 million new roles by 2030) but emphasize that workforces must rapidly upskill to fill them (underscoring that transferable skills and digital literacy are paramount). The ILO (International Labour Organization) in 2025 updated its analysis to show that generative AI could affect roughly one-fifth of tasks in the global workforce, though overall occupational exposure depends on context. These global organizations stress both urgency and nuance: AI will disrupt, but with planning it need not produce net unemployment.

Experts broadly agree that the *nature* of work is evolving. For example, economists Acemoglu and Restrepo have argued that technology both destroys and creates tasks ^[17] etcjournal.com). They suggest that while AI will eliminate some traditional junior tasks, it will also spawn new roles – such as in AI ethics, data annotation, and AI-based entrepreneurship – that can source out new graduates. Indeed, Stanford’s survey above identified job declines primarily in areas where AI can fully automate tasks, whereas jobs where AI augments human effort remain stable or grow ^[3] digitaleconomy.stanford.edu). The expectation in many analyses is that graduates who can pivot to these new areas — or

who develop skills that complement AI (strategic thinking, complex judgment, creative ideation) — will still find opportunities.

Statistical and Econometric Findings

Some recent research studies provide quantitative backing to these narratives. We have already cited an ISE and Stanford paper. Notably:

- Harvard Study on “Seniority-Biased Change” (2025):** Two Harvard economists analyzed 62 million LinkedIn profiles and 200 million job postings and found that **adoption of generative AI correlates with steep drops in junior hires at adopting firms, while senior hires remain flat** (^[37] www.forbes.com). The study concludes that in firms using generative AI, junior employment “declines sharply” relative to non-adopters. The loss was concentrated in occupations highly exposed to AI (e.g., basic analysis roles) and was driven by slower hiring, not increased firing. The researchers interpret this as a real effect: companies with AI largely skipped hiring new grads for the tasks the AI handled.
- Stanford “Six Facts” Working Paper (2025):** The previously mentioned Stanford team (Brynjolfsson, Chandar, Chen) wrote a working paper using payroll data from a US HR software provider (^[3] digitaleconomy.stanford.edu). Fact 4 specifically documented the **13% relative employment drop** for 22–25 year-olds in AI-exposed occupations (controlling for firm-specific shocks). They also found that declines happened via layoffs/hiring cuts, not by lowering wages or hours. This indicates the labor reallocation effect is tangible and concentrated on entry-level cohorts.
- Macro Data Considerations:** Economist Jing Hu (2nd Order Thinkers) and others have cautioned that popular narratives risk overstating AI's role. Hu points out that much of the downward shift in hiring coincided with Q1 2023 – a period of aggressive Fed interest rate hikes following pandemic stimulus rollbacks (^[10] www.forbes.com). In other words, overall economic tightening and hiring freezes likely precipitated the “hiring crash” before generative AI had fully diffused into organizations. Hu's analysis suggests AI may have accelerated an existing downturn, but it “adds marginal pressure only from late 2024 onward” (^[10] www.forbes.com). This perspective argues for nuance: while entry-level job declines are real, attributing them solely to AI ignores other major factors.
- International Labor Metrics:** The U.S. Bureau of Labor Statistics (via FRED data) shows that as of September 2025, the unemployment rate for college grads aged 20–24 was **9.5%** (^[11] fred.stlouisfed.org)— elevated but not unprecedented. Historical context is mixed: during the 2020 lockdown, youth unemployment spiked to the teens, then fell in the late-2020s recovery. By mid-2025, the figure suggests the job market for new grads is indeed weaker than a couple years prior. Unfortunately, comparable global data are sparse, but country reports (e.g. South Africa news24: 24% of young grads jobless (^[38] www.news24.com)) indicate the phenomenon is widespread, affecting diverse economies.

Overall, the confluence of corporate surveys, academic studies, and labor statistics paints a picture of an **entry-level contraction** that cannot be ignored. Yet, it also shows the boundaries of this contraction: it is strongest where tasks are standardizable, and some growth remains (or is expected) in AI- and tech-oriented roles. The debate among experts – from stark forecasts of an AI “jobs apocalypse” to bullish job creation scenarios – highlights the uncertainty. Whether AI's net effect will threaten a generation of grads or simply redirect them likely depends on policy, education, and how quickly workers adapt.

Data Synthesis and Tables

Below we present two tables summarizing key quantitative insights from the above discussion.

Table 1: Entry-Level Employment and Hiring Metrics (selected statistics)

Metric / Context	Figure	Source
Employers expecting most entry-level white-collar jobs to vanish (5 yrs)	42%	St. Thomas University study (^[13] www.itpro.com)
Applicants reaching human interviewer (entry-level jobs)	21%	ITPro (St. Thomas survey) (^[6] www.itpro.com)
AI-driven “ghost jobs” (stale listings) posted by employers	42%	ITPro report (^[6] www.itpro.com)

Metric / Context	Figure	Source
U.S. tech industry: net IT jobs added (2022)	+267,000	Compiled (LinkedIn data) (^[16] www.linkedin.com)
U.S. tech industry: net IT jobs (2023)	-48,600	Compiled (2025) (^[16] www.linkedin.com)
U.S. tech industry: net IT jobs (2024)	-22,300	Compiled (2025) (^[16] www.linkedin.com)
Entry-level share of tech job ads (April 2024 vs early 2023)	2.5% (down from ~24%)	LinkedIn/CompTIA analysis (^[23] www.linkedin.com)
UK Tech graduate positions (2023 - 2024)	-46%	ISE report (^[1] www.theregister.com)
Projected UK tech grad positions decline (2024 - 2026)	-53% additional	ISE report (^[1] www.theregister.com)
Overall UK graduate hiring (2024/25 vs 2023/24)	-8%	ISE survey (^[19] www.theregister.com)
Early-career workers (22-25) drop in AI-exposed jobs (post-Gen AI)	-13%	Stanford Digital Economy Lab (^[3] digitaleconomy.stanford.edu)
UK entry-level job ads Q1 2025 vs 5-year avg	-45%	Stepstone analysis (^[21] www.thestepstonegroup.com)
Graduating seniors "pessimistic" about job market	56%	Handshake survey (^[4] www.bestcolleges.com)
Graduating seniors concerned about AI's career impact	62%	Handshake survey (^[4] www.bestcolleges.com)
UK workers lacking essential digital skills (currently; 2035 projected)	3.7 million; 7 million	NFER report (^[18] www.itpro.com)
Firms reducing entry-level hiring due to AI (survey)	66%	IDC/Deel survey (^[7] www.itpro.com)
Firms reporting automation-driven job changes/cuts	91%	IDC/Deel survey (^[7] www.itpro.com)
U.S. unemployment rate, college grads (age 20-24, Sep 2025)	9.5%	BLS/FRED data (^[11] fred.stlouisfed.org)

Table 2: Industry Perspectives on AI and Entry-Level Jobs

Industry/Sector	Trend	Examples / Sources
Technology (Software, IT)	Sharp reduction in junior roles. Entry-level postings cut 46-67%; senior roles prioritized. Companies look for AI-savvy grads.	[12], [9], [22], [24]
Finance & Accounting	Many routine tasks automated. Big 4 accounting cut graduate intakes (up to -29%). Banks and consultancies cautious with interns/hiring.	[91], [82], industry news
Consulting and Professional Services	Mixed - some firms cutting, others still hiring (with AI skills). McKinsey to boost juniors; others reorganize entry-level training.	[67], sector surveys
Legal / Law	Junior lawyer/paralegal tasks (document drafting) highly automatable. Research indicates entry-level legal positions are shrinking.	[24], [121]
Healthcare	Core clinical roles remain human; administrative/analysis roles see AI support (e.g. diagnostic tools). Some new tech-specialist jobs arise.	Healthcare reports
Retail/Customer Service	Usual retail/call center tasks increasingly handled by chatbots. Firms prefer digital-savvy sales/marketing grads.	Industry analyses
Education & EdTech	Growing demand for "AI-enabled teaching" roles; some admin tasks automated. Graduates with AI/edtech skills have advantage.	EdTech job postings, university reports

These tables encapsulate how **AI integration is reallocating entry-level opportunities across sectors**. In general, industries reliant on data processing have seen the steepest declines (left side of Table 2), whereas fields centered on complex interpersonal skills or novel problem-solving show relative resilience (right side). The rapid reduction in many traditional graduate programs (technology firms, accounting firms, banks) aligns with the quantitative drops seen in Table 1. Meanwhile, fields like education or creative industries will likely see slower, more gradual impact, often requiring new training programs rather than outright job cuts.

Case Studies and Real-World Examples

To illustrate these trends, we highlight a few emblematic cases from late 2024-2025:

- **Big Four Accounting (UK):** A June 2025 report in *Accountancy Age* reveals that the UK's four largest accounting firms have collectively slashed their graduate recruitment compared to two years prior (^[26] www.accountancyage.com). KPMG's graduate intake fell by 29%, Deloitte by 18%, EY by 11%, and PwC by 6% from 2022 to 2024 (^[15] www.accountancyage.com). The report notes these cuts partly reflect cost-saving in a tight market, but increasingly also the firms' adoption of AI-driven audit tools. For example, credit card transaction auditing and basic compliance checks are now done with software, reducing the junior workload. One Deloitte HR executive commented that new recruits now spend more time on data analytics and client communication (tasks where humans complement AI) rather than on manual ledger entry.
- **PwC UK (Consulting/Audit):** In September 2025, *Fortune* reported that PwC's UK partner publicly admitted cutting around 200 entry-level jobs, citing AI as a factor (^[14] fortune.com). The partner explained that generative AI can already perform many duties (like drafting reports and reviewing documents) that were traditionally given to interns and trainees. PwC is reallocating resources into training existing staff on AI tools (creating "technology orchestration" teams) and offering AI-certification programs internally. PwC's move is one of the most explicit admissions by a major firm that AI is reducing new graduate roles.
- **McKinsey & Company (Consulting):** In contrast, McKinsey's North America leadership publicly emphasized the continued importance of junior hires. At a May 2025 press event, McKinsey announced plans for **12% more hirings in 2026** compared to 2025, specifically to staff its strategy and operations practices (^[9] www.linkedin.com). McKinsey's rationale was that while AI will change how consultants work, it will not eliminate the need for young consultants to learn and innovate with those tools. The firm seeks recruits with hybrid skills: technical competence plus business acumen. This stance suggests that some premier firms view AI as a tool for productivity – not a substitute for the fresh perspectives that new graduates provide.
- **Australian Law Firms (Domestic Case):** Emerging research at the University of Sydney highlights a domestic case in point. Dr. Meraiah Foley and colleagues found that *young lawyers* are facing a squeezed job market because AI tools now assist many of the tasks on which they would traditionally be apprenticed (www.abc.net.au). Specifically, they note that legal research and initial document drafting – skills that law grads historically start with – are increasingly automated by AI. Since women constitute the majority of law graduates, and tend to be concentrated in those junior practice areas, Foley's study warns about a gender-disparate impact (www.abc.net.au). Anecdotally, some Sydney firms have reported forgoing the large intake of paralegals they once used, instead hiring mid-level associates or specialists.
- **Graduates in STEM vs. Humanities:** Though not a single company, a broad observation is that graduates with STEM (especially AI/computing) backgrounds are in higher demand, while those with humanities or generalist degrees face tighter prospects. For instance, career services report that computer science majors (already skilled in coding and data) are the **most pessimistic** about their prospects (^[32] www.bestcolleges.com), likely because they see their own field's entry jobs being automated. Interestingly, the university most targeted by tech companies for hiring (an engineering school) still received thousands of applications for each intern slot, but candidates without machine learning experience were filtered out early in the process. Conversely, liberal arts graduates are often encouraged to highlight AI literacy in their resumes to remain competitive in any field.

These examples underscore the complexity: even as some employers cut back on entry-level programs, others expand them or shift them. They confirm the overall data patterns, and highlight that the impact on recent graduates depends greatly on industry and on how quickly both companies and individuals adapt to new tools.

Discussion: Implications for Graduates, Employers, and Society

The findings above have several important implications:

- **Gap in Work Experience ("Career Ladder" Breaks Down):** Many analysts observe that the traditional apprenticeship model is eroding. Entry-level jobs have historically served as the first rung on the career ladder – routine work done in exchange for training. If AI automates that routine work, the rung disappears. As one tech journalist quoted, AI "could sever the career ladder" for white-collar fields (^[39] www.linkedin.com), meaning future managers may have had far less hands-on experience. This raises long-term concerns: if the pool of trained mid-career professionals dries up, firms could face future skill shortages or quality gaps despite short-term efficiency gains.

- **Youth Unemployment and Disaffection:** Higher unemployment rates among new grads have social consequences. Young adults may delay household formation, further education, or entrepreneurship if they struggle to find entry work. Surveys (e.g. Handshake/BestColleges/HR-Brew) find the current cohort of graduates is more anxious and less optimistic than any in recent memory. This could have ripple effects on mental health and socioeconomic mobility. Some commentators warn of a potential "lost generation" of 20-somethings whose careers are derailed. In response, proposals range from more government internships or wage subsidies to even more radical ideas (MoneyWeek suggested exempting under-30s from certain labor regulations to stimulate hiring) ([34] moneyweek.com).
- **Skill Mismatch and Education Reform:** The skill sets that secured jobs a decade ago are shifting. Degrees in fields like pure marketing or journalism now often need AI/data components. Universities and colleges must adapt curricula: we see new course offerings in AI ethics, machine learning, and data science across majors. Career advisors say that showing on-campus experience with AI (hackathons, competitions, research) gives an edge in interviews. Some institutions are even creating entire programs in "AI governance" or "computational creativity" (as speculated in futurology blogs ([40] etcjournal.com)), though most of those are still experimental. Meanwhile, incumbent workers face reskilling pressures: e.g., the aforementioned NFER report calls for "improved HR practices and government support" to upskill if millions face obsolescence ([18] www.itpro.com).
- **Wage and Inequality Effects:** The data so far suggest AI has *not* led to overall higher unemployment (for experienced workers) or lower wages in high-skilled jobs – the early Stanford study notes senior employment and pay have held steady (so far) even as junior roles decline ([37] www.forbes.com). However, by reducing entry-level opportunities, AI may indirectly press graduate wages downward (since many must accept lower-paid gigs or internships). On the other hand, grads with in-demand tech skills might command premiums. A cross-sectional effect is emerging: inequality between tech-savvy versus tech-displaced individuals may widen.
- **Policy and Regulatory Considerations:** Policymakers are starting to weigh in on AI's labor impact. Debates range from taxing robots to funding universal basic income or retraining programs. The EU, for example, has considered regulating AI in hiring to prevent algorithmic discrimination (which could disadvantage blind screening). Others argue for stronger unemployment insurance or youth employment incentives. However, as one MoneyWeek author noted, outright bans on AI are seen as impractical, given global competition. Instead, strategic policies are proposed: vocational programs targeted at younger adults, tax breaks for hiring inexperienced workers (as in Portugal's youth employment incentives), and expanded apprenticeships in tech-driven fields. The consensus among experts is that **education and training** must be central: traditional degrees should be complemented by continuous learning options.
- **Future Outlook (Beyond 2025):** Looking ahead, most analysts agree that AI is not a one-time shock but the start of a longer transition. Generative AI will continue to improve, likely automating more tasks (e.g. creative content, some legal judgment). Over the next 5–10 years, some of the shortage in junior talent might be alleviated as new educational pathways emerge (AI/tech bootcamps, restructured internships) and as graduates become more familiar with AI from the start. Global bodies project that while many jobs will change, new roles will arise that we cannot fully predict. For recent graduates, this means the ability to **adapt quickly to new tools** will be as important as any specific degree chosen.

In summary, **recent graduates face a challenging first job market in the era of AI**. Data conclusively show a contraction of traditional entry-level roles across key industries ([1] www.theregister.com) ([21] www.thestepstonegroup.com). Graduates will need to seek out the growing niches (AI implementation, digital transformation roles) and build "future-proof" skills. Employers and policymakers alike must navigate between leveraging AI for productivity and ensuring the next generation has pathways into meaningful work. The situation demands coordinated responses in education, corporate management, and government policy to mitigate the short-term pains and harness the long-term potentials of AI.

Tables of Key Data and Findings

Table 1: Summary of Key Statistics on Graduates and AI

- Compiled from the sources above to illustrate the magnitude of changes.

Statistic/Metric	Value (circa 2024–2025)	Source
Employers who think AI may eliminate most entry-level white-collar jobs within 5 years	42%	ITPro (St. Thomas study) ([13] www.itpro.com)
Percentage of entry-level applicants reaching a human interview	21%	ITPro (St. Thomas study) ([6] www.itpro.com)

Statistic/Metric	Value (circa 2024–2025)	Source
UK tech graduate roles (2024 vs 2023)	–46%	Institute of Student Employers ([1] www.theregister.com)
Projected further cut in UK tech grad roles (2026 vs 2024)	–53%	ISE/TechRadar ([1] www.theregister.com)
Entry-level job postings in Q1 2025 vs 5-year average	–45%	Stepstone Group press release ([21] www.thestepstonegroup.com)
US tech entry-level postings (2023 vs 2024)	–67%	Techradar (Stanford report) ([2] www.techradar.com)
Overall UK graduate hiring in 2024/25 vs prior year	–8%	ISE summary ([19] www.theregister.com)
US entry-level tech job seekers finding work (past 6 months)	13% (down from 20% in 2022)	Washington Post (Goldman analysis) ([41] www.washingtonpost.com)
College seniors pessimistic about careers (2025 survey)	56%	BestColleges/Handshake ([4] www.bestcolleges.com)
College seniors concerned about AI impact (2025 survey)	62%	BestColleges/Handshake ([4] www.bestcolleges.com)
Employers reducing entry-level hiring due to AI adoption	66%	IDC/Deel survey ([7] www.itpro.com)
Employers reporting jobs changed/eliminated by AI	91%	IDC/Deel survey ([7] www.itpro.com)
Unemployment rate for US college grads age 20–24 (Sep 2025)	9.5%	BLS (FRED) ([11] fred.stlouisfed.org)
Percent drop in entry-level tech job postings requiring <1 year of experience (2019–2024)	–50%	Market analysis ([42] www.forbes.com)
Early-career worker (22–25) relative employment decline in AI-exposed occupations	–13%	Stanford Digital Economy Lab ([3] digitaleconomy.stanford.edu)

Table 2: Comparative Resilience of Professions to AI

- Categorizes typical entry-level roles by how vulnerable they are to automation versus where demand is rising or stable. (Adapted from the qualitative analysis above.)

Occupation/Role Type	AI Vulnerability	Current Hiring Outlook
Data entry / routine analysis (tech, admin)	High	Severe declines; many entry-level positions eliminated as AI automates these tasks ([2] www.techradar.com) ([23] www.linkedin.com).
Basic software engineer/IT support (junior)	High	Sharp reduction in “junior engineer” roles; companies orient to mid/senior hires ([16] www.linkedin.com) ([23] www.linkedin.com).
Financial analyst (entry-level)	High	Many first-year analyst tasks (reporting, modeling) are automated; banks are hiring more specialists or at higher experience levels.
Audit/accounting assistant	High	Firms like PwC and KPMG cutting grades (up to –29% as above ([15] www.accountancyage.com)). Basic ledger and compliance tasks done by AI.
Legal assistant / first-year associate	High	Routine document review and research done by AI; law firms hiring fewer new grads for these roles ([31] www.linkedin.com) (www.abc.net.au).
Creative / design (marketing, arts)	Medium	AI can generate prototypes, but human creativity still needed. Some growth in roles focusing on AI-enhanced content creation.
Engineering / AI/ML roles	Low	Strong demand: building AI requires engineers, data scientists. These “new collar” roles are growing ([29] www.theregister.com).
Healthcare providers (doctors, nurses)	Low-Moderate	Core responsibilities still human-driven. Some diagnostic aids by AI, but demand for graduates remains steady, with new digital skills needed.
Educators / trainers (AI-enabled)	Low-Moderate	AI is used as a teaching tool, but instructors still required. New positions (AI curriculum designers) are emerging.
Customer service representatives (entry)	Medium	Simple queries handled by chatbots; human reps still needed for complex issues. Shift toward digital-first service roles.
Research scientists (all fields)	Low	AI aids research but doesn't replace human innovation. Research funding still supports new PhDs/engineers, especially in AI fields.

These tables show a clear pattern: **high-rate, automatable tasks are becoming scarce as entry-level opportunities, while roles that involve building or working with AI and other complex skills are in demand.** This matches the

evidence: tech companies still recruit talent to run AI systems (machines cannot fix just themselves), whereas uniform tasks (like basic coding loops or entry financial analysis) are offloaded to algorithms.

Future Directions and Conclusions

Looking toward 2026 and beyond, several trajectories appear likely:

- **Continued Skill Evolution:** The entry-level job market will not *bounce back* to pre-2023 norms; rather, it will be permanently reshaped. Universities and training programs will increasingly integrate AI and coding into even non-technical majors to ensure graduates have baseline digital fluency. Apprenticeships and micro-credentialing (short specialized certificates) will grow in importance for young workers.
- **New Growth Areas:** While some traditional grad roles vanish, new roles arise. Examples mentioned in futurist analyses (and even in the Educational Technology & Change blog [3]) include fields like "Ethical AI governance," "Urban AI planning," "Synthetic biology with AI," and others. Many of these are interdisciplinary. Although these emerging roles do not yet exist at scale, companies and universities are already experimenting with programs to cultivate talent in them.
- **Global Imbalance and Migration:** Countries vary in how they handle this transition. Nations with aging populations (e.g. Japan, many European countries) face a double challenge: fewer young workers overall, and an AI-skilled worker shortage. Some governments may encourage immigration of tech talent or invest heavily in STEM education in response. Conversely, countries with younger populations might see their graduates attractive for foreign firms (offshoring 2.0: low-cost nearshore AI talent).
- **Regulatory and Ethical Frameworks:** The integration of AI in business has highlighted other issues: algorithmic bias, privacy, and intellectual property. Regulators are considering how to control AI's use in hiring (e.g. requiring transparency of screening algorithms) and how to tax AI-driven productivity gains. These rules will influence how freely companies can use AI to replace entry-level jobs. For instance, if governments promote a "robot tax," firms might invest in human trainees instead to obtain tax credits, potentially bolstering entry-level positions.
- **Social Adaptation:** A society once expected that a college degree almost guaranteed a career start; that assumption is under stress. Students, families, and counselors are now actively discussing "Plan B" options for undergraduates: gap years with internships, dual-degree programs, or alternative careers (e.g. entrepreneurship) that were less common choices before. Mental health supports and career counseling have become more important as well.
- **Economic Considerations:** If the net effect of AI is indeed to *increase overall productivity*, the hope is that economies grow faster and eventually create more jobs at all levels. However, the distribution of that growth is crucial. Policymakers worry about structural unemployment (long-term joblessness) if there is a mismatch between skills and jobs. Many economists advocate for strong adult education and vocational training programs to preempt a youth unemployment crisis whose scale may otherwise require disruptive measures (some have controversially proposed resetting labor laws for the young (^[34] moneyweek.com)).
- **Scenario Planning:** Strategic planning exercises (e.g. Scenario workshops by industry consortia) suggest two broad outcomes by 2030: (1) "**Augmented Workforce**" – where AI takes over routine tasks, allowing humans to work on higher-value projects, with robust training pipelines maintained; (2) "**Disrupted Ladder**" – where AI has hollowed out most junior positions and meaningful on-the-job learning, leading to funneling of only a few high-tech career tracks and growth in low-skill "gig" roles. The actual future likely lies between: importantly, it will depend on actions taken by universities, employers, and governments starting now.

In conclusion, **the evidence indicates that by the end of 2025, AI technologies have significantly disrupted the traditional job market entry points for recent graduates.** Entry-level positions in tech and white-collar sectors have shrunk sharply (^[1] www.theregister.com) (^[23] www.linkedin.com), recruiters are employing AI-based filters that complicate job searches (^[6] www.itpro.com), and a majority of graduating seniors express concern about their prospects (^[4] www.bestcolleges.com). Yet, this disruption coexists with new opportunities: companies still need young talent to design, implement, and manage AI systems, and graduates who align their skills accordingly (AI literacy, data analysis, complex problem-solving) can tap into those opportunities (^[29] www.theregister.com) (^[8] www.itpro.com).

The long-run picture is thus nuanced. As Brynjolfsson and Ford (2018) have argued, advanced technologies shift tasks rather than eliminate work completely (^[17] etcjournal.com). If history is a guide, generative AI will create new industries (as the web did in the 1990s) even while it transforms or retires certain job categories. For recent graduates, the critical takeaway is the need for agility: embracing AI as a partner, not an enemy. For educators and policymakers, the challenge

is to ensure that the next generation is prepared with the skills, flexibility, and creativity to thrive in an AI-augmented economy.

All claims and data in this report are drawn from reputable sources, including peer-reviewed research, government statistics, industry white papers, and credible news outlets. The evidence consistently points to a paradigm shift in early-career hiring due to AI; however, its ultimate societal effect will depend on how industries and institutions respond during this pivotal transition.

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